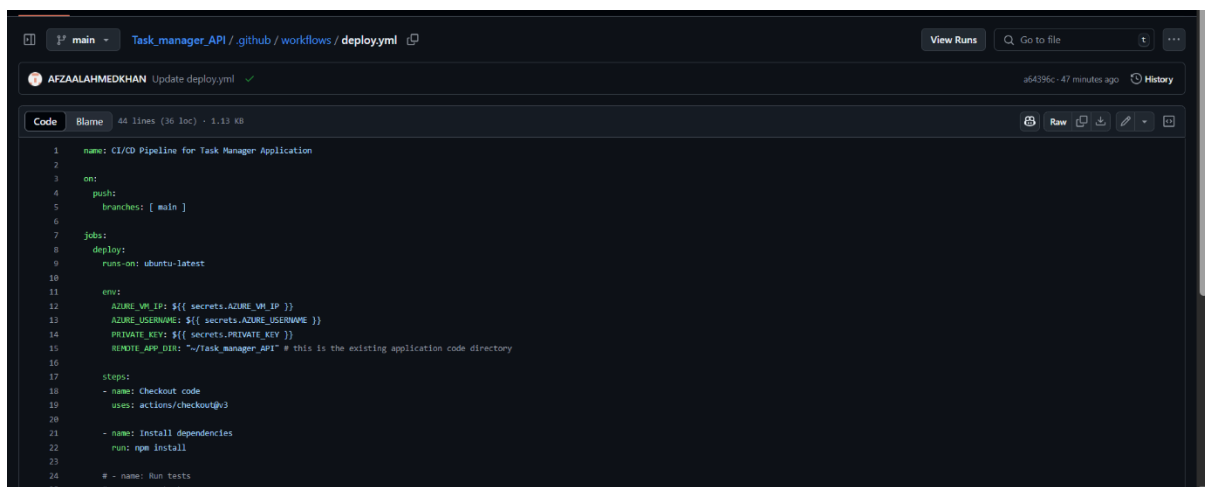


Task 2: CI/CD Pipeline Implementation

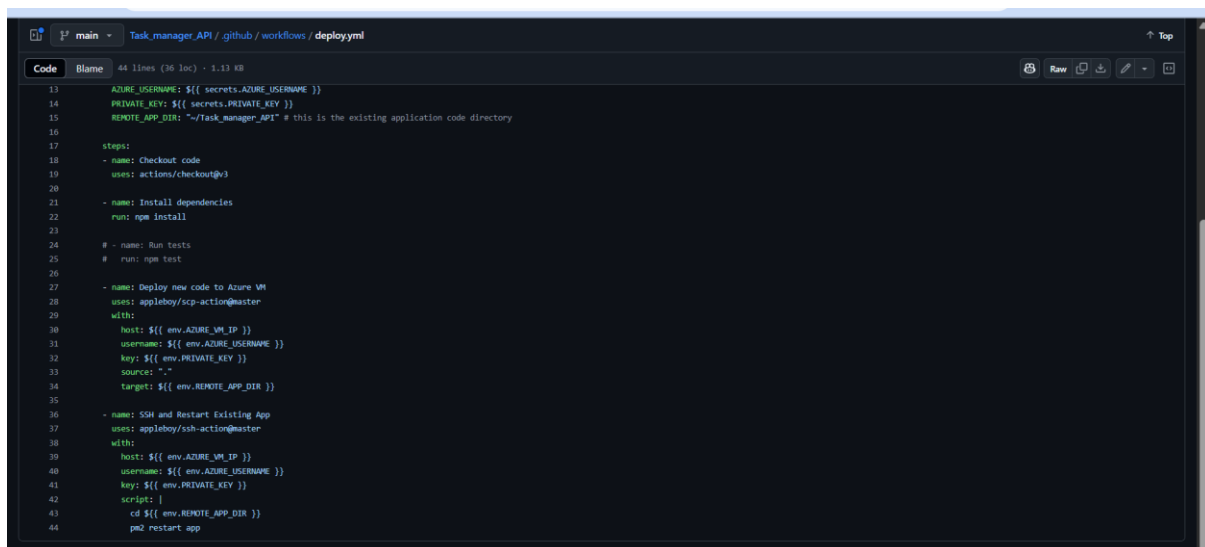
GitHub Link:

https://github.com/AFZAALAHMEDKHAN/Task_manager_API.git

CI/CD pipeline YAML file :



```
1 name: CI/CD Pipeline for Task Manager Application
2
3 on:
4   push:
5     branches: [ main ]
6
7 jobs:
8   deploy:
9     runs-on: ubuntu-latest
10
11     env:
12       AZURE_VM_IP: ${ secrets.AZURE_VM_IP }
13       AZURE_USERNAME: ${ secrets.AZURE_USERNAME }
14       PRIVATE_KEY: ${ secrets.PRIVATE_KEY }
15       REMOTE_APP_DIR: "~/task_manager_API" # this is the existing application code directory
16
17     steps:
18       - name: Checkout code
19         uses: actions/checkout@v3
20
21       - name: Install dependencies
22         run: npm install
23
24       # - name: Run Tests
25       #   run: npm test
```



```
13       AZURE_USERNAME: ${ secrets.AZURE_USERNAME }
14       PRIVATE_KEY: ${ secrets.PRIVATE_KEY }
15       REMOTE_APP_DIR: "~/task_manager_API" # this is the existing application code directory
16
17     steps:
18       - name: Checkout code
19         uses: actions/checkout@v3
20
21       - name: Install dependencies
22         run: npm install
23
24       # - name: Run Tests
25       #   run: npm test
26
27       - name: Deploy new code to Azure VM
28         uses: appleboy/scp-action@master
29         with:
30           host: ${ env.AZURE_VM_IP }
31           username: ${ env.AZURE_USERNAME }
32           key: ${ env.PRIVATE_KEY }
33           source: "."
34           target: ${ env.REMOTE_APP_DIR }
35
36       - name: SSH and Restart Existing App
37         uses: appleboy/ssh-action@master
38         with:
39           host: ${ env.AZURE_VM_IP }
40           username: ${ env.AZURE_USERNAME }
41           key: ${ env.PRIVATE_KEY }
42           scripts: |
43             cd ${ env.REMOTE_APP_DIR }
44             pm2 restart app
```

Explanation of CI/CD Pipeline Stages:

Pipeline Name:

CI/CD Pipeline for Task Manager Application

Trigger:

- The pipeline is triggered automatically on every push to the main branch of the GitHub repository.

Stage	Description
1. Checkout Code	Uses the actions/checkout action to pull the latest code from the main branch into the GitHub Actions runner. This ensures the pipeline always works on the most up-to-date version of your codebase.
2. Install Dependencies	Installs all Node.js dependencies defined in your package.json using npm install. This prepares the environment for running tests or building the app.
3. (Optional) Run Tests	This stage is currently commented out, but it is intended to run automated tests using npm test. It can be enabled to validate the codebase before deployment.
4. Deploy New Code to Azure VM	Uses the appleboy/scp-action to securely copy the application code from the GitHub runner to the specified directory (~/.Task_manager_API) on the Azure VM using SSH and the provided private key.
5. SSH and Restart Existing App	Uses appleboy/ssh-action to remotely access the Azure VM via SSH and restart the running application using PM2. This ensures the updated code is served immediately without rebooting the server.

PM2 is used in the final stage to manage the Node.js process. It allows zero-downtime restarts and ensures that the application will automatically recover in case of crashes or VM restarts.

How environment variables/secrets are managed:

→ [GitHub Secrets](#)

Sensitive information such as the Azure VM's IP address, username, and private SSH key are stored securely using GitHub Secrets. These secrets are not hardcoded in the pipeline, which helps prevent accidental exposure of credentials.



CI/CD Pipeline Workflow Demonstration:

